Global System Rules (The "Chain of Custody")

These rules apply across all four faces of the platform to ensure safety and data integrity:

- **Internal Rating Engine:** All quality ratings (thumbs up/down, text reviews) are visible only to the Central NGO backend, never to the public. This prevents "food bank shaming" while allowing the algorithm to downrank low-quality donors or poorly managed centers secretly.
- **Geospatial Verification:** Every QR code scan automatically logs the GPS Lat/Long of the device to prevent fraud and create an auditable chain of custody.
- **Visual Quality Management:** Require a photo of the food at three distinct lifecycle stages: at the event (Donor), at pickup (Driver), and at drop-off (Distribution Center).

Face 1: The Consumer (The Needy)

Platform: Mobile-first web app.
Goal: Dignified discovery and dietary-safe matching.

- **Profile & Dietary Matching:** Users select FDA-recognized allergens from a dropdown. An LLM parses any custom text input (e.g., "I avoid high sodium") into structured JSON to filter available centers.
- **Navigation:** Integrates Google Maps API to show transit routes to centers that currently have food matching their dietary profile.
- **Interaction:** Users scan the Distribution Center's entrance QR code.
- **Feedback Loop:** Upon leaving, users give a simple Thumbs Up/Down and optional text review. The backend LLM parses the text to determine if the complaint was about the venue (Face 2) or a specific food item/donor (Face 3).

Face 2: Distribution Centers (Soup Kitchens / Shelters)

Platform: Mobile-first web app.
Goal: Inventory receipt and facility profiling.

- **Facility Profiling:** Centers register their operating hours, seating, appliances (refrigerators, warmers), and staffing levels.
- **Intake & Verification:** When a driver arrives, the driver scans the Center's unique "Location QR." The Center staff then take the final verification photo of the arriving food.
- **Feedback Loop:** Staff can submit direct reviews on specific food batches (via QR scan) to warn the NGO about spoiled deliveries or inappropriate packaging.

Face 3: Event Organizers (The Donors)

Platform: Mobile-first web app/ also accessible through website/ native software
Goal: Frictionless donation and pre-planning.

- **Pre-Event Ingestion:** Organizers snap a photo of their catering menu/invoice. An LLM OCRs the document, extracts dishes, ingredients, and allergens, and allows human verification. This data alerts the NGO to plan logistics days in advance.
- **Post-Event Triage:** The organizer opens the pre-loaded menu, selects what is left over, estimates servings, and takes the "Stage 1" photo.
- **Predictive Logistics:** If the human forgets to specify transport needs, the LLM infers it from the food type (e.g., "Ice cream detected -> Requires deep freeze transport").

Face 4: The Central NGO (Logistics & Routing)

Platform: Backend Engine + Driver Mobile View + website/local admin dashboard
Goal: AI-driven dispatch and system orchestration.

- **The Driver Workflow:** The driver arrives at the event, scans the Event QR, takes the "Stage 2" pickup photo, and assigns a unique QR tracking sticker to the physical food container. Drivers carry a roll of **pre-printed, blank QR stickers**. When they pick up the food, they stick a random QR code on the box, scan it with the app, and the backend permanently links that random QR code to the specific digital food batch.

- **Algorithmic Routing:** The backend calculates the most efficient drop-off route based on driver location, food lifespan (hot vs. cold), and the registered appliances of nearby Distribution Centers (Face 2).
- **The LLM Brain:** The central system ingests condensed JSON payloads from all faces. It manages the priority queue, punishes bad actors based on parsed reviews, and updates the real-time map for Consumers (Face 1).

HACKATHON SHORTCUT: "Mock" Face 1 (The Consumer). Building four separate frontends in React will kill your team's velocity.

- **The Fix:** Build Face 2 (Center), Face 3 (Donor), and Face 4 (Driver/NGO Map) as fully functioning web apps. For Face 1 (The Consumer), build a single, static mobile UI mockup just to show the judges the "vision" of how the needy interact with the system, but do not wire it up to the database. The true technical innovation is in the logistics (Faces 3 and 4).